

HOUSE No. 3543

The Commonwealth of Massachusetts

PRESENTED BY:

Steven Owens

To the Honorable Senate and House of Representatives of the Commonwealth of Massachusetts in General Court assembled:

The undersigned legislators and/or citizens respectfully petition for the adoption of the accompanying bill:

An Act relative to the protection and development of the thermal commons of the Commonwealth.

PETITION OF:

NAME:	DISTRICT/ADDRESS:	DATE ADDED:
<i>Steven Owens</i>	<i>29th Middlesex</i>	<i>1/17/2025</i>

HOUSE No. 3543

By Representative Owens of Watertown, a petition (accompanied by bill, House, No. 3543) of Steven Owens relative to thermal commons of the Commonwealth. Telecommunications, Utilities and Energy.

The Commonwealth of Massachusetts

In the One Hundred and Ninety-Fourth General Court
(2025-2026)

An Act relative to the protection and development of the thermal commons of the Commonwealth.

Be it enacted by the Senate and House of Representatives in General Court assembled, and by the authority of the same, as follows:

1 SECTION 1. Chapter 25 of the General Laws, as so appearing, is hereby amended by
2 adding after Section 23 the following section:-

3 Section 24, (a) As used in this section, the following words shall have the following
4 meanings unless the context clearly requires otherwise:-

5 “Ambient geothermal energy”, geothermal energy of less than 80 degrees Fahrenheit
6 found at depths of 2,000 feet or less in the earth.

7 “Anthropogenic geothermal energy”, geothermal energy of less than 80 degrees
8 Fahrenheit found at depths of 2000 feet or less and originating from human-caused climate
9 change.

10 "Geothermal energy", thermal energy that is sourced from (i) the ground, including
11 bedrock and the earth beneath the bedrock, (ii) the surface water, including the rivers, ponds,

lakes within the commonwealth and the sea adjacent to the commonwealth, and (iii) the water beneath the surface of the earth within the commonwealth, including groundwater, springs, and aquifers. Such geothermal energy shall include the electricity derived therefrom.

“Hot geothermal energy”, geothermal energy of more than 80 degrees Fahrenheit found at depths of 10,000 feet or more in the earth .

“Waste Thermal Energy”, activities by humans that produce thermal energy as a byproduct and include thermal energy from sources including, but not limited to, wastewater, industrial sources, landfills, and data centers.

(b) It is hereby declared that ambient geothermal energy and anthropogenic geothermal energy within the commonwealth as defined herein are a thermal commons held for the benefit of the citizens of the commonwealth in public trust, except where found on property wholly owned by private property owners or where produced wholly on private land, including but not limited to waste thermal energy.

(c) To ensure that the commonwealth achieves its mandate to reduce greenhouse gas emissions to net zero by the year 2050, the secretary of the executive office of energy and environmental affairs shall establish a commission/working group to investigate and make recommendations to the legislature for a comprehensive, equitable and efficient framework to manage the opportunities presented to the commonwealth by the thermal energy transition from combusting to non-combusting sources of thermal energy. Such commission shall be called the Commission/Working Group for the Protection and Development of the Commonwealth’s Thermal Energy Commons.

(d) The commission/working group shall consist of 20 members, including 1 person representing each of the following: MAPC & other regional councils, energy efficiency experts, thermal energy experts, utility experts, organized labor, EJ organization, AGO, DEP, DOER, MassCEC, Executive office of economic development, and Massachusetts Bar Association with expertise in land use. Interested parties shall apply to the secretary for designation as members. The commissioner of the department of public utilities shall serve as chair of the commission/working group.

(e) The commission/working group shall review available research and data and shall define the scope of the thermal energy derived from the commons of the commonwealth. It shall further solicit and consider relevant comments from stakeholders and shall develop recommendations for legislative and regulatory changes to facilitate the protection and development for the benefit of the citizens of the commonwealth the thermal commons, with due consideration given to land use impacts and categorizations, water quality, soil health, and restoration of and maintenance of the thermal stability of the commonwealth. The commission/working group shall work with the executive office of energy and environmental affairs to gather information and data to quantify how protection and development of the thermal energy commons of the commonwealth may contribute to meeting the greenhouse gas emission reductions requirements of chapter 21N. The recommendations of the commission/working group shall include, but not be limited to, the following areas of proposed legislation or regulation: (i) the extent of the thermal commons including but not limited to setbacks from the boundaries of any private or public property, differences in usage requirements between anthropogenic and non-anthropogenic ambient geothermal energy, (ii) any obligation to provide essential energy service in return for a grant of access to the thermal commons, and the resulting

prioritization and regulatory framework of such thermal service; (iii) establish a reciprocal obligation to serve the public in return for a grant of access to the thermal commons; (iv) to establish regulatory priorities for such service to the public based on the provision of essential energy services, safety, security, reliability, equity, emissions and affordability; (v) the scope of permission for the drawdown of anthropogenic ambient geothermal energy, which shall not be lower in annual average temperatures than experienced by the commonwealth in the year 1900; (vi) the extent to which drawdown of anthropogenic geothermal energy shall be considered restoration of the thermal commons and compensated for such restoration); (vii) to encourage drawdown of anthropogenic geothermal energy as an ecosystem restoration: (viii) to ensure annual stability of non-anthropogenic ambient geothermal energy through appropriate design and management of the thermal commons.

(f) The commission/working group shall conduct no fewer than three public hearings in conveniently accessible locations throughout the commonwealth. The executive office of energy and environmental affairs shall provide administrative support for the operations of the commission/working group. The commission/working group may retain independent expert consultants.

(g) The commission/working group shall submit a report and recommendations, together with any drafts of legislation that may be useful in carrying out its recommendations and otherwise putting them into effect to the department of public utilities, the joint committee on telecommunications, utilities and energy, the senate and house committees on global warming and climate change and the clerks of the senate and house of representatives not later than July 31, 2026.